Phase 1: Recruiter Portal & Assessment Creation
Recruiter Login / JWT Auth (Admin approval gate: is_approved)
Job Posting Creation (Title, Dept, Description / JD PDF)
Upload Candidate Resume (PDF / DOCX)
DocumentExtractor Utility (Extracts Raw Text from JD & Resume)
Groq AI ATS Screening Service (llama-3-70b / gpt-oss-20b) Extracts Skills, Experience & Highlights
ATS Match Score Calculated (0 - 100%)
Score & Skill Matrix Ready
Configure Interview Parameters (num_questions=5, start_date, start_time)
InviteCandidateView Backend
- Generates UUID Session Token
- Creates 12-char Temp Password
- Creates Candidate User (is_first_login=True)
- link1: 48h Expirable Candidate Link
- link2: Permanent Recruiter Tracking Link
SMTP Email Service (Sends HTML Invite Email with Magic Link)
In-App Notifications (Candidate Created & Email Sent)
Magic Link Access
Phase 2: Candidate Onboarding & Pre-Flight Checks
Candidate Clicks Magic Link (/interview/:sessionToken)
Password Reset / JWT Authentication (Token verification & session resolution)
6-Stage Pre-Flight System Check (SystemCheck.jsx)
1. Camera Feed Verification
2. Mic Verification (Singleton AudioContext & Level Meter)
3. Screen Share Stream Acquisition
4. Network Ping & Bandwidth Latency Test
5. Legal Biometric & Consent Agreement
6. Room & Lighting Check
No
All 6 Hardware Checks Passed?
Yes
Waiting Room (WaitingRoom.jsx) Synchronized Countdown to Slot Time
Slot Arrives / Session Starts
Phase 3: Real-Time Adaptive Interview & 3D Evaluation Loop
Interview Chamber Mounts (Interview.jsx)
- Fullscreen Lock Enabled
- Establishes Channels WebSocket (/ws/interview/:token/)
- Establishes STT WebSocket (/ws/stt/:token/)
Monitored Concurrently
GetInterviewSessionView (Resolves interview or rehydrates state from F5 reload)
GroqAIService.generate_next_question()
Generates Foundational Question #0
Appends to question_bank & Broadcasts
Adaptive Q&A Execution Loop (Questions 0 to 4)
AI Question Spoken & Displayed (Concise <25 words, Human Conversational Tone)
Continuous Audio Ingestion
- Singleton AudioContext & ScriptProcessorNode
- 16kHz Mono Linear PCM Chunks
Speech-to-Text Routing Engine
- Deepgram Nova-3 (Cloud Low-Latency)
- Groq Whisper-large-v3
- Gemini Live Bidi / Faster-Whisper
Real-time Transcript Finalized
Candidate Submits Verbal / Text Answer
Forces Early Finish
Yes: Next Question
SubmitAnswerView API (Receives question_index, question_text, answer_text)
Parallel ThreadPoolExecutor (Concurrent Processing)
Thread B:
ai_service.generate_next_question()
- Incorporates JD + Resume Claims
- Ingests Full Conversation History
- Strict Prohibited Question List (Zero Repeat)
- Non-IT Role Boundary Enforcement
- Adaptive Difficulty Scaling:
* >85: Senior / Architecture Deep-Dive
* 60-85: Intermediate Tooling / Projects
* <60: Foundational Recovery Pivot
Thread A:
ai_service.evaluate_answer()
3-Dimensional Scoring:
1. Relevance (0-100)
2. Accuracy (0-100)
3. Clarity (0-100)
+ Overall Score & Feedback
Persist Response Model in PostgreSQL (question_index, relevance, accuracy, clarity)
Channels ASGI Layer Broadcast (group_send 'answer_event' & 'question_event')
WebSocket Event Push
Question Index < 4?
No: 5 Questions Finished
Phase 4: Multi-Layered Proctoring & Anomaly Engine
Proctoring Listener (Document Visibility State)
- Tab Switch Detection
- Window Blur Event
- Fullscreen Exit Attempt
Dual Stream Canvas Blending
Captures Webcam Frame + Screen Share Frame
LogAnomalyView Backend
Saves AnomalyLog with Severity (low/medium/high)
Anomaly Severity > Threshold
or Malpractice Detected?
Yes: Malpractice
Flag Malpractice Status
- Terminate Interview Immediately
- Force Overall Score = 0
- Enforce Recommendation = Reject
Real-time Anomaly Push
Phase 5: Recruiter Live Monitoring Command Center
Recruiter Live Monitoring Portal (LiveMonitoring.jsx)
Connected via WebSockets
/ws/interview/:token/
Real-Time Dashboard Updates:
- Live Q&A Stream
- Instant 3D Scores (Rel/Acc/Clar)
- Proctoring Anomaly Alerts & Snapshots
- Candidate Live Status
Phase 6: Final Evaluation Synthesis & Comprehensive Reporting
Candidate Review Form (Feedback.jsx)
- 1 to 5 Star Ratings (Experience, AI Clarity, Ease, Stability)
- Qualitative Comments
Asynchronous Report Worker (generate_interview_report)
- Aggregates All 5 Q&A Responses
- Groq Llama-3 Generates Overall Score (0-100)
- Generates Executive Summary
- Compiles Verified Strengths & Weaknesses
- Generates Actionable Hiring Recommendation (Hire/Reject/Maybe)
Persist Report in Database & Local/Supabase Storage (Stores Report JSON & Updates Screening Status)
Financial Telemetry & INR Cost Engine
- Live USD-to-INR API Integration
- Computes Token Inference Cost in ₹ (₹10-15 per candidate)
Recruiter Candidate Scorecard (ReportDetail.jsx)
- Full Transcript & 3D Radar Breakdown
- Strengths vs Weaknesses Matrix
- Anomaly Timeline Audit
- Export Verified PDF Hiring Scorecard
Recruiter Final Decision (Shortlist / Reject / On Hold)